

GEPA

Reflective prompt evolution (Agrawal et al., ICLR 2026)

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ETH Zurich

6 October 2026

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Section 1

The problem

GSM8K: grade-school maths word problems

Q: Alannah, Beatrix, and Queen are preparing for the new school year and have been given books by their parents. Alannah has 20 more books than Beatrix. Queen has $\frac{1}{5}$ times more books than Alannah. If Beatrix has 30 books, how many books do the three have together?

Yang et al. 2024, Fig. 3 and section 5

GSM8K: grade-school maths word problems

Q: Alannah, Beatrix, and Queen are preparing for the new school year and have been given books by their parents. Alannah has 20 more books than Beatrix. Queen has $\frac{1}{5}$ times more books than Alannah. If Beatrix has 30 books, how many books do the three have together?

The answer on file

140

Yang et al. 2024, Fig. 3 and section 5

GSM8K: grade-school maths word problems

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A: The instruction goes here Then the working, then a number

At the start of the answer

The answer on file

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Yang et al. 2024, Fig. 3 and section 5

GSM8K: grade-school maths word problems

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Right if
they match

Yang et al. 2024, Fig. 3 and section 5

GSM8K: grade-school maths word problems

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The answer on file

140

Right if
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At the start of the answer



7,473 training problems
To optimise on (a 3.5% sample)

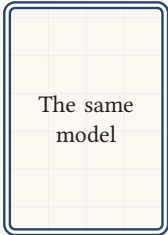


1,319 test problems
To check the result on

Yang et al. 2024, Fig. 3 and section 5

Same model, different instruction

Accuracy on the GSM8K test set

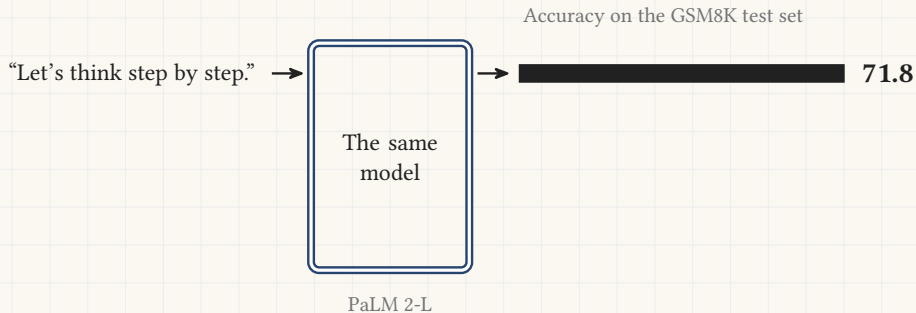


The same
model

PaLM 2-L

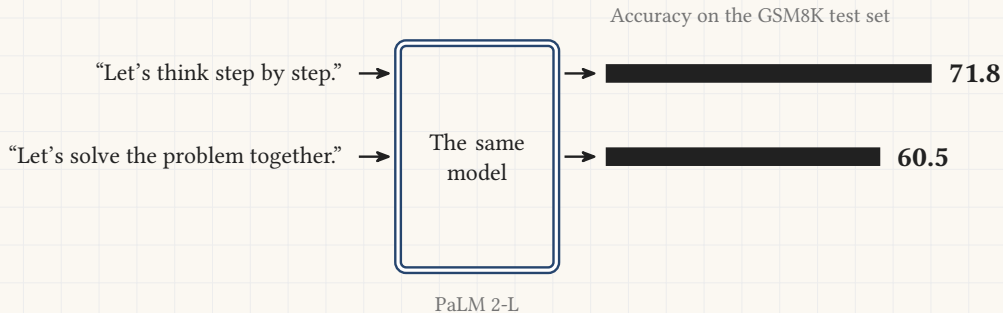
Yang et al. 2024, section 5.2.3, p. 13

Same model, different instruction



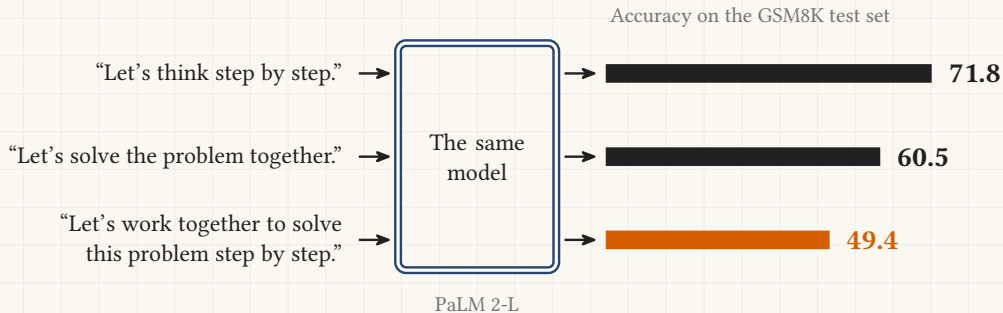
Yang et al. 2024, section 5.2.3, p. 13

Same model, different instruction



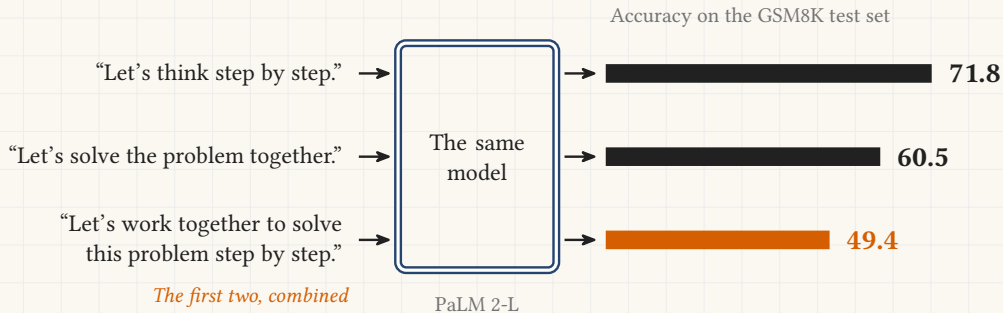
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Same model, different instruction



Yang et al. 2024, section 5.2.3, p. 13

One instruction, before and after GEPA

The seed instruction

```
create_query_hop2
```

Given the fields
question, summary_1,
produce the fields query.

Agrawal et al. 2026, Fig. 2, Table 2

One instruction, before and after GEPA

The seed instruction

`create_query_hop2`

Given the fields
question, summary_1,
produce the fields query.

GEPA's instruction, GPT-4.1 Mini

`create_query_hop2`

[...] **Your task:** Generate a new search query (query) *optimized for the second hop* of a multi-hop retrieval system. [...]

Key Observations and Lessons:

- First-hop documents often cover one entity or aspect.
- Remaining relevant documents often involve connected or higher-level concepts mentioned in summary_1 but not explicitly asked in the original question. [...]
- If summary_1 describes a population for a small civil parish, but the question wants the total population of the wider region, your query should target that wider region (e.g., “Madeira archipelago population in 2011”). [...]

Agrawal et al. 2026, Fig. 2, Table 2

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Lessons written by
the reflecting model

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Lessons written by
the reflecting model

System's test score,
HotpotQA, GPT-4.1 Mini

Seed		38.00
GEPA		69.00

GEPA's instruction, GPT-4.1 Mini

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Agrawal et al. 2026, Fig. 2, Table 2

The optimisation problem

An input

$$\Phi(\overset{\text{An input}}{\underset{\text{all instructions; all weights}}{x; \langle \Pi, \Theta \rangle}})$$

Π : all instructions; Θ : all weights

$\underbrace{\hspace{10em}}$
Output y of the system Φ

Agrawal et al. 2026, eq. (1) and (2)

The optimisation problem

Metric, a score in $[0, 1]$

An input Gold answer, a rubric or unit tests

$$\mu\left(\Phi\left(\underset{\text{Output } y \text{ of the system } \Phi}{x; \langle \Pi, \Theta \rangle}\right), m\right)$$

Π : all instructions; Θ : all weights

Agrawal et al. 2026, eq. (1) and (2)

The optimisation problem

Metric, a score in $[0, 1]$

An input

Gold answer, a rubric or unit tests

$$\mathbb{E} \left[\mu \left(\Phi(x; \underbrace{\langle \Pi, \Theta \rangle}_{\text{Output } y \text{ of the system } \Phi}), m \right) \right]$$

Π : all instructions; Θ : all weights

Output y of the system Φ

Over (x, m) drawn from the task distribution

Agrawal et al. 2026, eq. (1) and (2)

The optimisation problem

$\langle \Pi^*, \Theta^* \rangle$ = the instructions and weights that maximise

Metric, a score in $[0, 1]$

$$\mathbb{E} \left[\mu \left(\Phi \left(\underbrace{x}_{\text{An input}}, \underbrace{\langle \Pi, \Theta \rangle}_{\text{Gold answer, a rubric or unit tests}} \right), m \right) \right] \quad \text{subject to} \quad \# \text{rollouts} \leq \underbrace{B}_{\text{Budget}}$$

Π : all instructions; Θ : all weights

Output y of the system Φ

A rollout: one run of Φ on one input, scored by μ .

Over (x, m) drawn from the task distribution

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$\langle \Pi^*, \Theta^* \rangle$ = the instructions and weights that maximise

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Output y of the system Φ

Over (x, m) drawn from the task distribution

A rollout: one run of Φ on one input, scored by μ .

GEPA: weights frozen, instructions searched

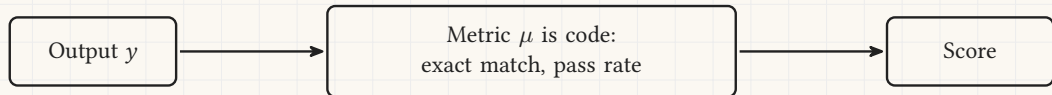
Agrawal et al. 2026, eq. (1) and (2)

2

Section 2

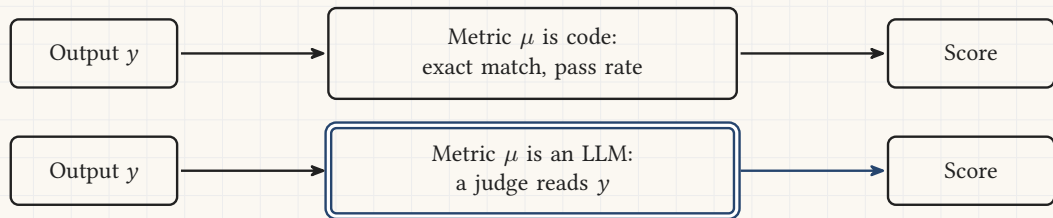
Learning from text

The metric can be an LLM



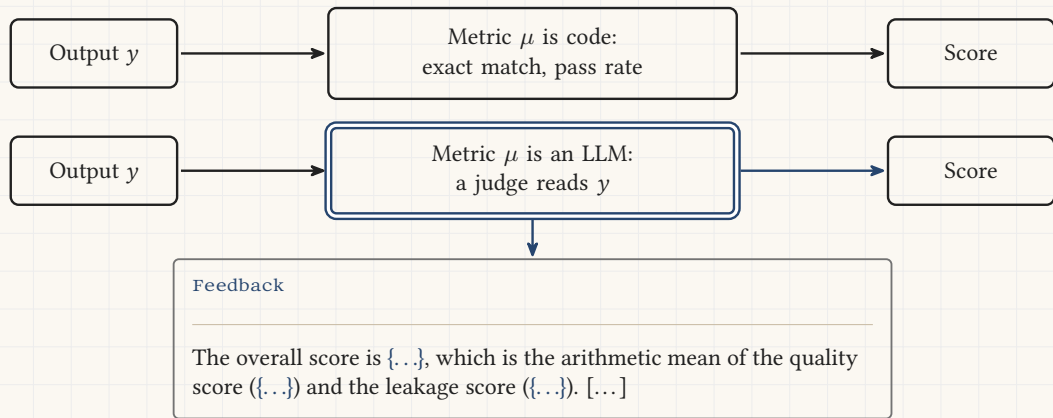
Agrawal et al. 2026, pp. 4, 5, 25; DSPy tutorial “GEPA for Privacy-Conscious Delegation”

The metric can be an LLM



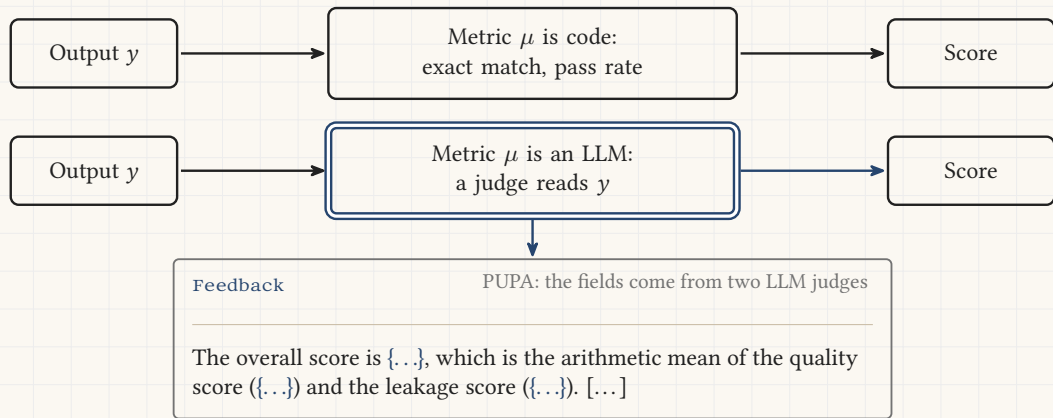
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A rollout returns a score and text

`system_prompt`

The seed instruction

You are a helpful assistant. You are given a question and you need to answer it. The answer should be given at the end of your response in exactly the format
'### <final answer>'

GEPA's recorded AIME test; Agrawal et al. 2026, pp. 6, 7

A rollout returns a score and text

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The seed instruction

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Problem x (2022 AIME I, problem 6)

Find the number of ordered pairs of integers (a, b) such that the sequence 3, 4, 5, a , b , 30, 40, 50 is strictly increasing and no set of four (not necessarily consecutive) terms forms an arithmetic progression.

GEPA's recorded AIME test; Agrawal et al. 2026, pp. 6, 7

A rollout returns a score and text

system_prompt

The seed instruction

You are a helpful assistant. You are given a question and you need to answer it. The answer should be given at the end of your response in exactly the format '### <final answer>'

ooo

Execution trace of GPT-4.1 nano, last line

<final answer> 234

Problem x (2022 AIME I, problem 6)

Find the number of ordered pairs of integers (a, b) such that the sequence 3, 4, 5, a , b , 30, 40, 50 is strictly increasing and no set of four (not necessarily consecutive) terms forms an arithmetic progression.

Gold answer m : ### 228

GEPA's recorded AIME test; Agrawal et al. 2026, pp. 6, 7

A rollout returns a score and text

system_prompt

The seed instruction

You are a helpful assistant. You are given a question and you need to answer it. The answer should be given at the end of your response in exactly the format '### <final answer>'

ooo

Execution trace of GPT-4.1 nano, last line

<final answer> 234

Metric μ : is ### 228 in the response? No: score 0

Problem x (2022 AIME I, problem 6)

Find the number of ordered pairs of integers (a, b) such that the sequence 3, 4, 5, a , b , 30, 40, 50 is strictly increasing and no set of four (not necessarily consecutive) terms forms an arithmetic progression.

Gold answer m : ### 228

GEPA's recorded AIME test; Agrawal et al. 2026, pp. 6, 7

A rollout returns a score and text

system_prompt

The seed instruction

You are a helpful assistant. You are given a question and you need to answer it. The answer should be given at the end of your response in exactly the format '### <final answer>'

Problem x (2022 AIME I, problem 6)

Find the number of ordered pairs of integers (a, b) such that the sequence 3, 4, 5, a , b , 30, 40, 50 is strictly increasing and no set of four (not necessarily consecutive) terms forms an arithmetic progression.

Gold answer m : ### 228

GEPA's recorded AIME test; Agrawal et al. 2026, pp. 6, 7

ooo

Execution trace of GPT-4.1 nano, last line

<final answer> 234

Metric μ : is ### 228 in the response? No: score 0

Feedback function μ_f returns the score and this text:

Feedback

Evaluation trace: what the metric wrote

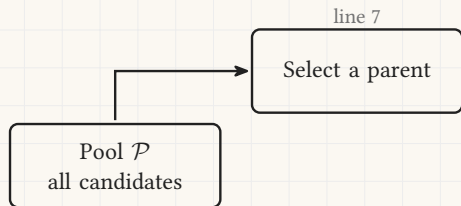
The generated response is incorrect. The correct answer is '### 228'. Ensure that the correct answer is included in the response exactly as it is. Here is some additional context that might be helpful: solution: [...]

3

Section 3

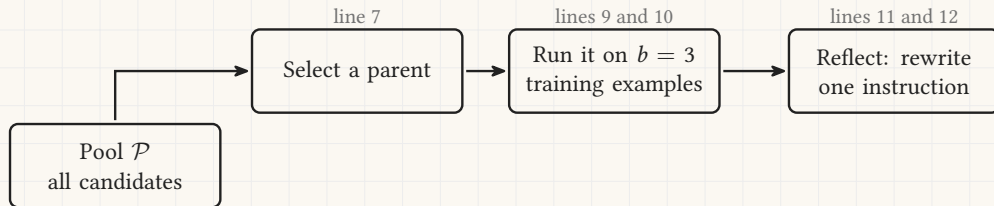
The GEPA loop

GEPA's loop, with the lines of Algorithm 1



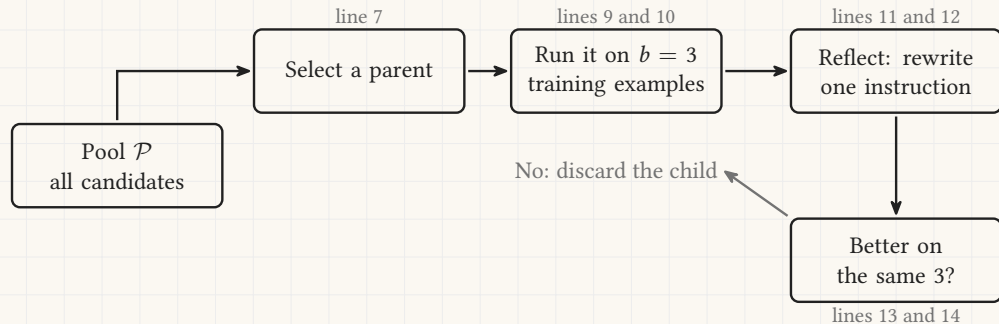
Agrawal et al. 2026, Fig. 3 and Alg. 1

GEPA's loop, with the lines of Algorithm 1



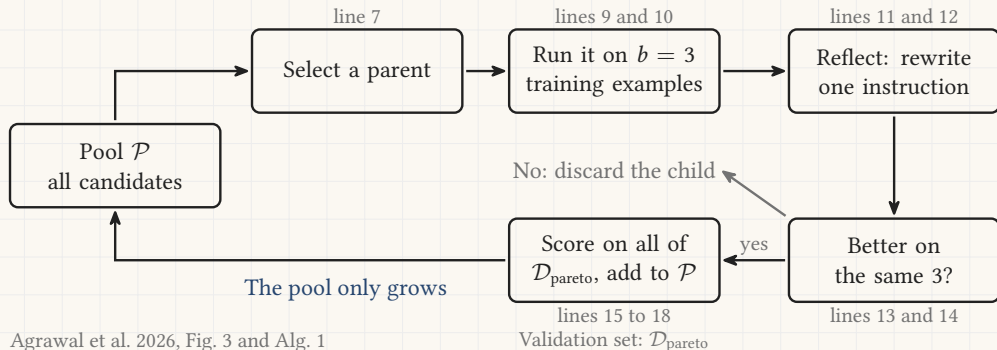
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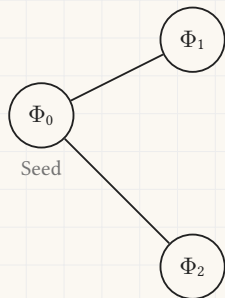
GEPA grows a tree of candidates



Seed

Agrawal et al. 2026, Alg. 1 and Fig. 6

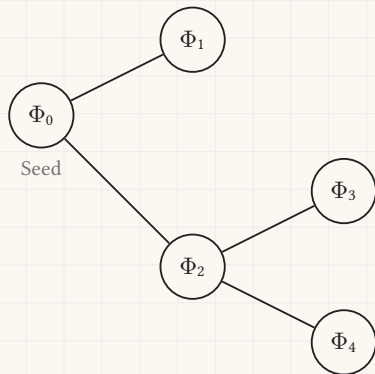
GEPA grows a tree of candidates



Each child is its parent with one instruction rewritten after reflection.

Agrawal et al. 2026, Alg. 1 and Fig. 6

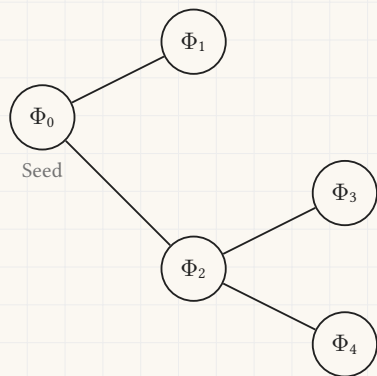
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Agrawal et al. 2026, Alg. 1 and Fig. 6

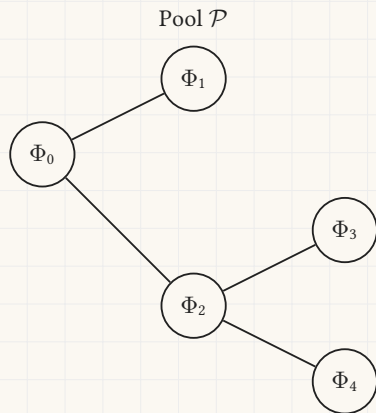
GEPA grows a tree of candidates



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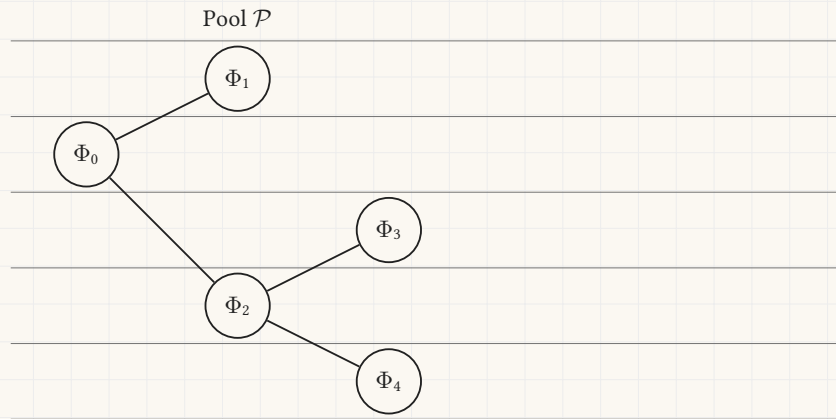
Agrawal et al. 2026, Alg. 1 and Fig. 6

The Pareto front of the pool



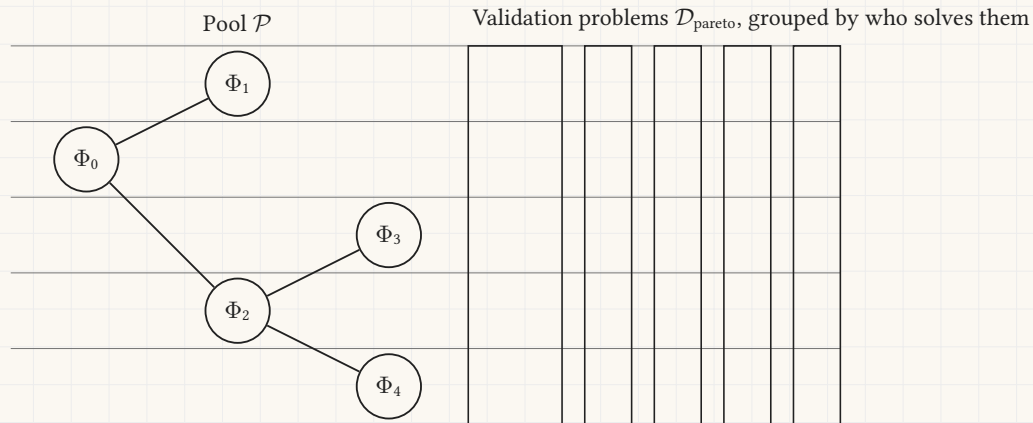
Agrawal et al. 2026, p. 7 and Alg. 2

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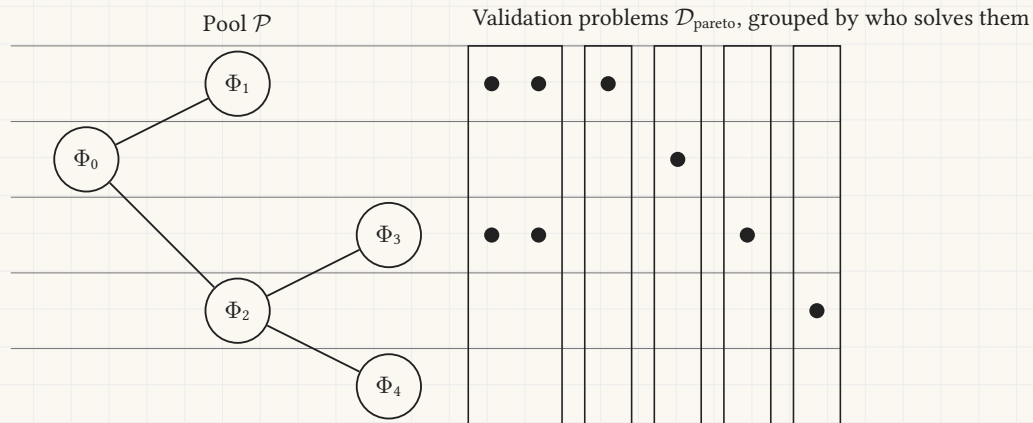
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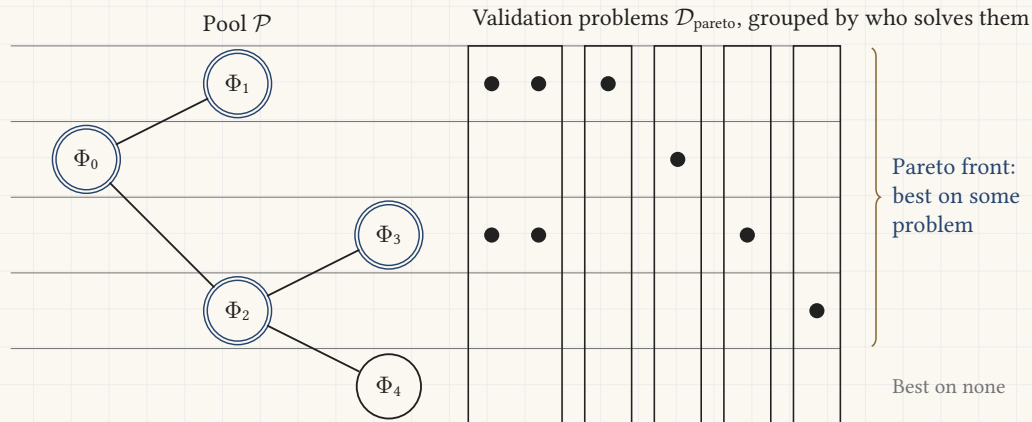
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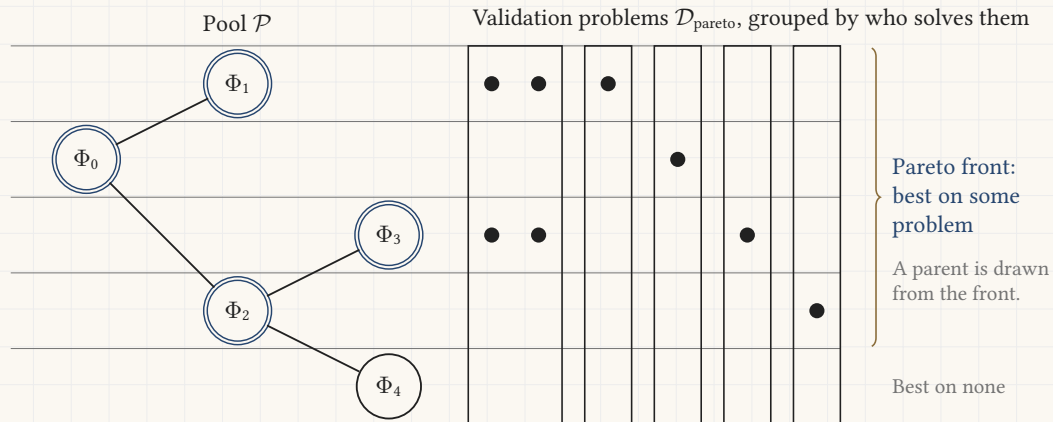
Agrawal et al. 2026, p. 7 and Alg. 2

The Pareto front of the pool



Agrawal et al. 2026, p. 7 and Alg. 2

The Pareto front of the pool



Agrawal et al. 2026, p. 7 and Alg. 2

Setup

HotpotQA, questions
 $\mathcal{D}_{\text{feedback}}$
Training split 150

Minibatches of $b = 3$
Reflection reads inputs, outputs, feedback

Agrawal et al. 2026, Alg. 1, pp. 8, 23, 27

Setup

	HotpotQA, questions		
$\mathcal{D}_{\text{feedback}}$		150	Minibatches of $b = 3$
Training split			Reflection reads inputs, outputs, feedback
$\mathcal{D}_{\text{pareto}}$		300	Each accepted child runs on all
Validation split			Scores only, to select candidates

Agrawal et al. 2026, Alg. 1, pp. 8, 23, 27

Setup

	HotpotQA, questions		
$\mathcal{D}_{\text{feedback}}$ Training split		150	Minibatches of $b = 3$ Reflection reads inputs, outputs, feedback
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Test Held out		300	Reported once, at the end

Agrawal et al. 2026, Alg. 1, pp. 8, 23, 27

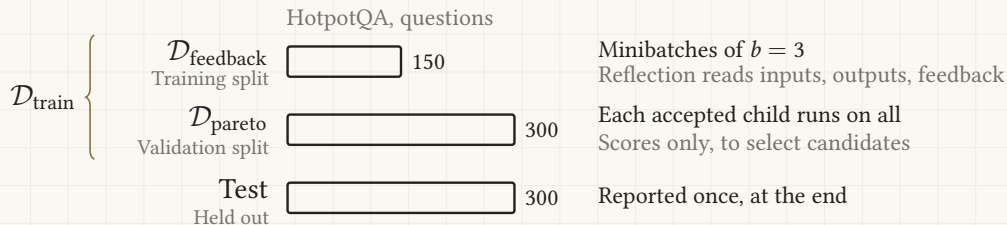
Setup

HotpotQA, questions		
$\mathcal{D}_{\text{train}}$ {	$\mathcal{D}_{\text{feedback}}$ Training split	150
	$\mathcal{D}_{\text{pareto}}$ Validation split	300
	Test Held out	300
		Minibatches of $b = 3$ Reflection reads inputs, outputs, feedback
		Each accepted child runs on all Scores only, to select candidates
		Reported once, at the end

- 1 **Split $\mathcal{D}_{\text{train}}$ into $\mathcal{D}_{\text{feedback}}$, $\mathcal{D}_{\text{pareto}}$, s.t. $|\mathcal{D}_{\text{pareto}}| = n_{\text{pareto}}$**
- 2 Initialize candidates $\mathcal{P} \leftarrow [\Phi]$
- 3 **for each (x_i, m_i) in $\mathcal{D}_{\text{pareto}}$ do**
- 4 $S_{\Phi}[i] \leftarrow \mu(\Phi(x_i), m_i)$

Agrawal et al. 2026, Alg. 1, pp. 8, 23, 27

Setup



- 1 Split $\mathcal{D}_{\text{train}}$ into $\mathcal{D}_{\text{feedback}}$, $\mathcal{D}_{\text{pareto}}$, s.t. $|\mathcal{D}_{\text{pareto}}| = n_{\text{pareto}}$
- ▶ 2 **Initialize candidates** $\mathcal{P} \leftarrow [\Phi]$ *// Φ : the seed*
- ▶ 3 **for each** (x_i, m_i) **in** $\mathcal{D}_{\text{pareto}}$ **do** *// all 300 questions*
- ▶ 4 $S_{\Phi}[i] \leftarrow \mu(\Phi(x_i), m_i)$ *// S_{Φ} : one row of scores*

Agrawal et al. 2026, Alg. 1, pp. 8, 23, 27

Algorithm 1 in words

Algorithm 1, GEPA

- ▶ **while the budget B is not exhausted do**
 - ▶ **$\Phi \leftarrow$ select a candidate from the pool $\mathcal{P}$**
 - $\mathcal{M} \leftarrow$ sample a few questions from $\mathcal{D}_{\text{feedback}}$
 - Use Φ to solve $\mathcal{M}$. Collect feedback F , score σ , traces
 - $\Phi' \leftarrow$ update Φ with F , σ and the traces
 - Get the score σ' of Φ' on $\mathcal{M}$
 - if $\sigma' > \sigma$ then**
 - Add Φ' to $\mathcal{P}$ as a child of Φ

Algorithm 1 in words

Algorithm 1, GEPA

```
while the budget  $B$  is not exhausted do
   $\Phi \leftarrow$  select a candidate from the pool  $\mathcal{P}$ 
  ▶  $\mathcal{M} \leftarrow$  sample a few questions from  $\mathcal{D}_{\text{feedback}}$ 
  ▶ Use  $\Phi$  to solve  $\mathcal{M}$ . Collect feedback  $F$ , score  $\sigma$ , traces
   $\Phi' \leftarrow$  update  $\Phi$  with  $F$ ,  $\sigma$  and the traces
  Get the score  $\sigma'$  of  $\Phi'$  on  $\mathcal{M}$ 
  if  $\sigma' > \sigma$  then
    Add  $\Phi'$  to  $\mathcal{P}$  as a child of  $\Phi$ 
```

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 ► **$\Phi' \leftarrow$ update Φ with F, σ and the traces**
 Get the score σ' of Φ' on $\mathcal{M}$
 if $\sigma' > \sigma$ **then**
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 ▶ **if $\sigma' > \sigma$ then**
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Algorithm 1 in words

Algorithm 1, GEPA

while the budget B is not exhausted **do**
 $\Phi \leftarrow$ select a candidate from the pool $\mathcal{P}$
 $\mathcal{M} \leftarrow$ sample a few questions from $\mathcal{D}_{\text{feedback}}$
 Use Φ to solve $\mathcal{M}$. Collect feedback F , score σ , traces
 $\Phi' \leftarrow$ update Φ with F , σ and the traces
 Get the score σ' of Φ' on $\mathcal{M}$
 if $\sigma' > \sigma$ **then**
 ► Add Φ' to $\mathcal{P}$ as a child of Φ

Algorithm 1: one iteration

Algorithm 1, GEPA Agrawal et al. 2026, p. 6; GEPA library

```
6 while budget  $B$  not exhausted do                                     //  $B$  in rollouts
▶ 7    $k \leftarrow \text{SELECTCANDIDATE}(\mathcal{P}, S)$                              // draw a parent
▶ 8    $j \leftarrow \text{SELECTMODULE}(\Phi_k)$                                      // which instruction, round robin
9    $\mathcal{M} \leftarrow$  minibatch of size  $b$  from  $\mathcal{D}_{\text{feedback}}$ 
10  Gather feedback, scores, traces for  $\Phi_k[j]$  on  $\mathcal{M}$  using  $\mu_f$ 
11   $\pi'_j \leftarrow \text{UPDATEPROMPT}(\pi_j, \text{feedbacks}, \text{traces}[j])$ 
12   $\Phi' \leftarrow$  Copy of  $\Phi_k$  w/ module  $j$  updated by  $\pi'_j$ 
13   $\sigma, \sigma' \leftarrow$  avg score on  $\mathcal{M}$  (before, after)
14  if  $\sigma'$  improved then
15    Add  $\Phi'$  to  $\mathcal{P}$ ; Add  $k$  to  $\mathcal{A}$ 
16    for each  $(x_i, m_i)$  in  $\mathcal{D}_{\text{pareto}}$ :  $S_{\Phi'}[i] \leftarrow \mu(\Phi'(x_i), m_i)$ 
21 return  $\Phi^*$  maximizing average score on  $\mathcal{D}_{\text{pareto}}$ 
```

Algorithm 1: one iteration

Algorithm 1, GEPA Agrawal et al. 2026, p. 6; GEPA library

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6 while budget  $B$  not exhausted do //  $B$  in rollouts
7    $k \leftarrow \text{SELECTCANDIDATE}(\mathcal{P}, S)$  // draw a parent
8    $j \leftarrow \text{SELECTMODULE}(\Phi_k)$  // which instruction, round robin
▶ 9    $\mathcal{M} \leftarrow$  minibatch of size  $b$  from  $\mathcal{D}_{\text{feedback}}$  //  $b = 3$  training examples
▶ 10  Gather feedback, scores, traces for  $\Phi_k[j]$  on  $\mathcal{M}$  using  $\mu_f$  // parent rerun, with traces
11   $\pi'_j \leftarrow \text{UPDATEPROMPT}(\pi_j, \text{feedbacks}, \text{traces}[j])$ 
12   $\Phi' \leftarrow$  Copy of  $\Phi_k$  w/ module  $j$  updated by  $\pi'_j$ 
13   $\sigma, \sigma' \leftarrow$  avg score on  $\mathcal{M}$  (before, after)
14  if  $\sigma'$  improved then
15    Add  $\Phi'$  to  $\mathcal{P}$ ; Add  $k$  to  $\mathcal{A}$ 
16    for each  $(x_i, m_i)$  in  $\mathcal{D}_{\text{pareto}}$ :  $S_{\Phi'}[i] \leftarrow \mu(\Phi'(x_i), m_i)$ 
21 return  $\Phi^*$  maximizing average score on  $\mathcal{D}_{\text{pareto}}$ 
```

Algorithm 1: one iteration

Algorithm 1, GEPA Agrawal et al. 2026, p. 6; GEPA library

```
6 while budget  $B$  not exhausted do                                     //  $B$  in rollouts
7    $k \leftarrow \text{SELECTCANDIDATE}(\mathcal{P}, S)$                                // draw a parent
8    $j \leftarrow \text{SELECTMODULE}(\Phi_k)$                                    // which instruction, round robin
9    $\mathcal{M} \leftarrow$  minibatch of size  $b$  from  $\mathcal{D}_{\text{feedback}}$              //  $b = 3$  training examples
10  Gather feedback, scores, traces for  $\Phi_k[j]$  on  $\mathcal{M}$  using  $\mu_f$          // parent rerun, with traces
▶ 11   $\pi'_j \leftarrow \text{UPDATEPROMPT}(\pi_j, \text{feedbacks}, \text{traces}[j])$     // an LLM rewrites instruction  $\pi_j$ 
▶ 12   $\Phi' \leftarrow \text{Copy of } \Phi_k \text{ w/ module } j \text{ updated by } \pi'_j$  // the child
13   $\sigma, \sigma' \leftarrow$  avg score on  $\mathcal{M}$  (before, after)
14  if  $\sigma'$  improved then
15    Add  $\Phi'$  to  $\mathcal{P}$ ; Add  $k$  to  $\mathcal{A}$ 
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12    $\Phi' \leftarrow$  Copy of  $\Phi_k$  w/ module  $j$  updated by  $\pi'_j$          // the child
▶ 13    $\sigma, \sigma' \leftarrow$  avg score on  $\mathcal{M}$  (before, after)           // same  $b$  examples
▶ 14   if  $\sigma'$  improved then                                           // strictly better
▶ 15     Add  $\Phi'$  to  $\mathcal{P}$ ; Add  $k$  to  $\mathcal{A}$                                      //  $\mathcal{P}$ : pool;  $\mathcal{A}$ : parents
16     for each  $(x_i, m_i)$  in  $\mathcal{D}_{\text{pareto}}$ :  $S_{\Phi'}[i] \leftarrow \mu(\Phi'(x_i), m_i)$ 
21  return  $\Phi^*$  maximizing average score on  $\mathcal{D}_{\text{pareto}}$ 
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15     Add  $\Phi'$  to  $\mathcal{P}$ ; Add  $k$  to  $\mathcal{A}$                                      //  $\mathcal{P}$ : pool;  $\mathcal{A}$ : parents
16     for each  $(x_i, m_i)$  in  $\mathcal{D}_{\text{pareto}}$ :  $S_{\Phi'}[i] \leftarrow \mu(\Phi'(x_i), m_i)$  //  $S_{\Phi'}$ : all validation scores
17     return  $\Phi^*$  maximizing average score on  $\mathcal{D}_{\text{pareto}}$              // best average
```

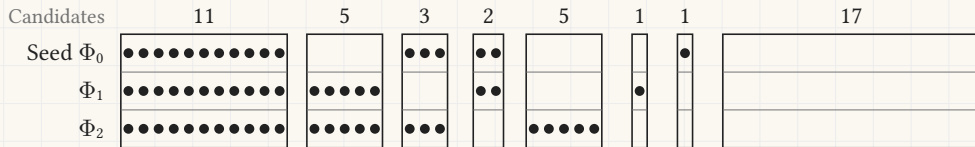
4

Section 4

Choosing the parent

The score matrix of a real run

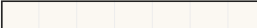
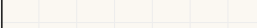
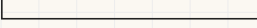
After iteration 5: 45 validation problems, grouped by who solves them



DSPy tutorial “GEPA for AIME”: GPT-4.1 Mini solves, GPT-5 reflects

The score matrix of a real run

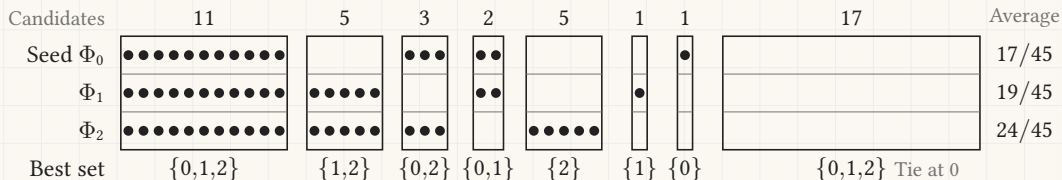
After iteration 5: 45 validation problems, grouped by who solves them

Candidates	11	5	3	2	5	1	1	17	Average
Seed Φ_0									17/45
Φ_1									19/45
Φ_2									24/45

DSPy tutorial “GEPA for AIME”: GPT-4.1 Mini solves, GPT-5 reflects

The score matrix of a real run

After iteration 5: 45 validation problems, grouped by who solves them



Φ is best on problem i when
no candidate scores higher on i

DSPy tutorial “GEPA for AIME”: GPT-4.1 Mini solves, GPT-5 reflects

The score matrix of a real run

After iteration 5: 45 validation problems, grouped by who solves them

Candidates	11	5	3	2	5	1	1	17	Average
Seed Φ_0									17/45
Φ_1									19/45
Φ_2									24/45
Best set	{0,1,2}	{1,2}	{0,2}	{0,1}	{2}	{1}	{0}	{0,1,2} Tie at 0	

Φ is best on problem i when
no candidate scores higher on i

Dominance: as high on every
problem, and higher on one

Each alone best somewhere:

Φ_2 on 5, Φ_1 on 1, Φ_0 on 1

DSPy tutorial “GEPA for AIME”: GPT-4.1 Mini solves, GPT-5 reflects

Algorithm 2 in words

Algorithm 2, Pareto-based candidate selection

► **function** SELECTCANDIDATE($\mathcal{P}$)

► $\mathcal{C} \leftarrow \emptyset$

► **for each** problem q in the validation set $\mathcal{D}_{\text{pareto}}$ **do**

► **Add to** $\mathcal{C}$ **all** Φ **in the pool** $\mathcal{P}$ **that best answer** q

Remove from $\mathcal{C}$ any Φ dominated by another. What is left: $\hat{\mathcal{C}}$

For Φ in $\hat{\mathcal{C}}$, let $f(\Phi)$ be the number of problems where Φ gives the best answer

Sample Φ from $\hat{\mathcal{C}}$ with probability proportional to $f(\Phi)$

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- ▶ **For Φ in $\hat{\mathcal{C}}$, let $f(\Phi)$ be the number of problems where Φ gives the best answer**
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Algorithm 2 on the matrix

Algorithm 2, Pareto-based candidate selection Agrawal et al. 2026; DSPy tutorial

```
1  function SELECTCANDIDATE( $\mathcal{P}$ ,  $S$ )                                     // the run's pool, scores
▶ 3    for each  $i$  do                                                    // i: 45 validation problems
▶ 4         $s^*[i] \leftarrow \max_k S_{\mathcal{P}[k]}[i]$                                 // top score on  $i$ 
▶ 5         $\mathcal{P}^*[i] \leftarrow \{\mathcal{P}[k] : S_{\mathcal{P}[k]}[i] = s^*[i]\}$         // best sets, shown before
7     $\mathcal{C} \leftarrow$  unique candidates in  $\bigcup_i \mathcal{P}^*[i]$ 
8     $D \leftarrow \emptyset$ 
9    while there exists  $\Phi \in \mathcal{C} \setminus D$  dominated by another in  $\mathcal{C} \setminus D$  do
10        $D \leftarrow D \cup \{\Phi\}$ 
12    Remove  $D$  from each  $\mathcal{P}^*[i]$  to get  $\hat{\mathcal{P}}^*[i]$ 
13    Let  $f[\Phi] =$  number of  $i$  for which  $\Phi \in \hat{\mathcal{P}}^*[i]$ 
14    Sample  $\Phi_k$  from  $\hat{\mathcal{C}}$  with probability  $\propto f[\Phi_k]$ 
15    return index  $k$  of  $\Phi_k$  in  $\mathcal{P}$ 
```

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▶ 7       $\mathcal{C} \leftarrow$  unique candidates in  $\bigcup_i \mathcal{P}^*[i]$                 // Pareto front:  $\Phi_0, \Phi_1, \Phi_2$ 
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▶ 8     $D \leftarrow \emptyset$                                             // D: the dominated
▶ 9    while there exists  $\Phi \in \mathcal{C} \setminus D$  dominated by another in  $\mathcal{C} \setminus D$  do // none here
▶ 10    $D \leftarrow D \cup \{\Phi\}$                                      //  $\hat{\mathcal{C}} = \mathcal{C} \setminus D$ 
12   Remove  $D$  from each  $\mathcal{P}^*[i]$  to get  $\hat{\mathcal{P}}^*[i]$ 
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▶ 13   Let  $f[\Phi] =$  number of  $i$  for which  $\Phi \in \hat{\mathcal{P}}^*[i]$              //  $f = 34, 36, 41$ 
14   Sample  $\Phi_k$  from  $\hat{\mathcal{C}}$  with probability  $\propto f[\Phi_k]$ 
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▶ 14    Sample  $\Phi_k$  from  $\hat{\mathcal{C}}$  with probability  $\propto f[\Phi_k]$            // 34/111, 36/111, 41/111
▶ 15    return index  $k$  of  $\Phi_k$  in  $\mathcal{P}$                                // run drew  $\Phi_1$ , then  $\Phi_2$ 
```

GEPA against GRPO and MIPROv2, per task

Qwen3 8B, test score (%)

	HotpotQA	IFBench	HoVer	PUPA	AIME 2025	LiveBench Math	Aggregate
Baseline	42.33	36.90	35.33	80.82	27.33	48.70	45.23

Agrawal et al. 2026, Table 1; Opsahl-Ong et al. 2024

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Baseline	42.33	36.90	35.33	80.82	27.33	48.70	45.23
GRPO	43.33	35.88	38.67	86.66	38.00	51.26	48.91
Rollouts, GRPO	24,000	24,000	24,000	24,000	24,000	24,000	

Agrawal et al. 2026, Table 1; Opsahl-Ong et al. 2024

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Baseline	42.33	36.90	35.33	80.82	27.33	48.70	45.23
GRPO	43.33	35.88	38.67	86.66	38.00	51.26	48.91
MIPROv2	55.33	36.22	47.33	81.55	20.00	46.60	47.84
Instructions and examples, chosen by Bayesian search							
Rollouts, GRPO	24,000	24,000	24,000	24,000	24,000	24,000	

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Instructions and examples, chosen by Bayesian search							
GEPA	62.33	38.61	52.33	91.85	32.00	51.95	54.85
Rollouts, GRPO	24,000	24,000	24,000	24,000	24,000	24,000	
Rollouts, GEPA	6,871	3,593	7,051	2,426	1,839	1,839	

Agrawal et al. 2026, Table 1; Opsahl-Ong et al. 2024

GEPA against GRPO and MIPROv2, per task

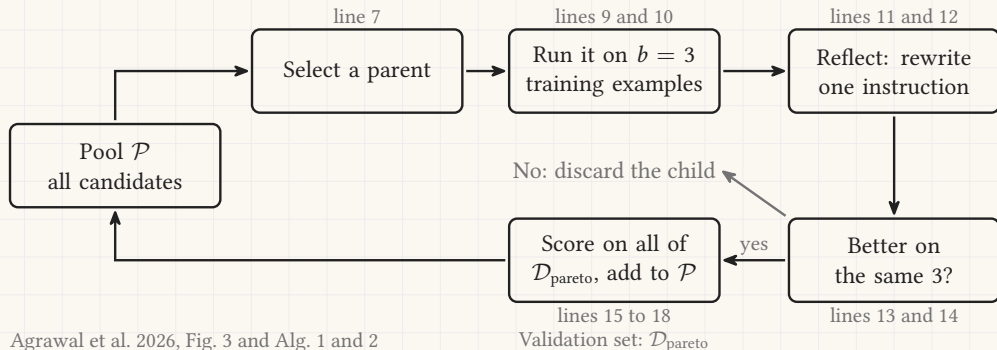
Qwen3 8B, test score (%)

GRPO ahead: 38.00 against 32.00

	HotpotQA	IFBench	HoVer	PUPA	AIME 2025	LiveBench Math	Aggregate
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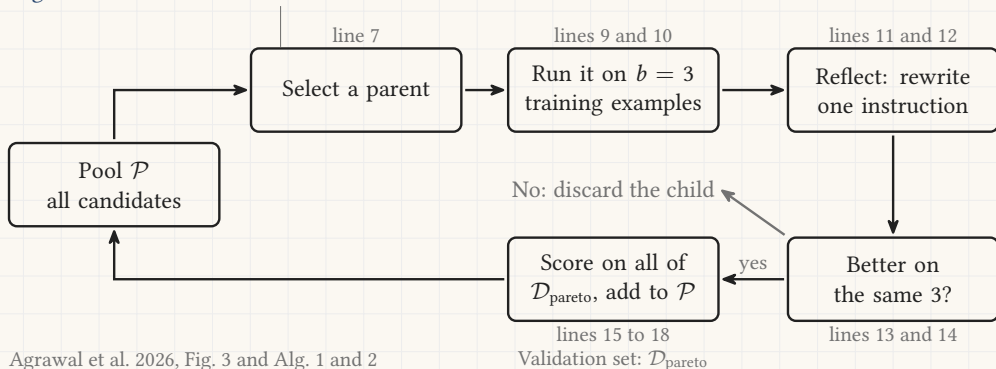
GEPA, assembled



Agrawal et al. 2026, Fig. 3 and Alg. 1 and 2

GEPA, assembled

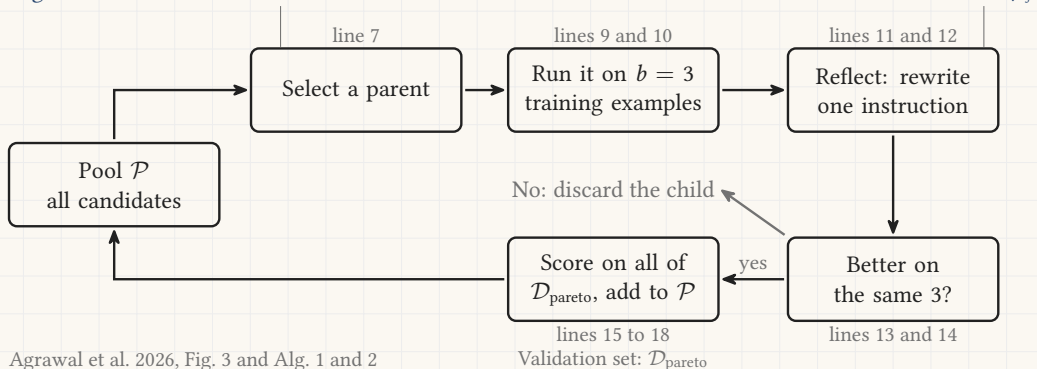
Algorithm 2: a draw from the Pareto front



Agrawal et al. 2026, Fig. 3 and Alg. 1 and 2

GEPA, assembled

Algorithm 2: a draw from the Pareto front



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Runs quoted in this lecture: the GEPA library's recorded AIME test, `tests/test_aime_prompt_optimization` at commit `d771eb2`; the DSPy tutorials “GEPA for AIME”, a run of 12 August 2025, and “GEPA for Privacy-Conscious Delegation”, dspy.ai/tutorials/gepa_aime and dspy.ai/tutorials/gepa_papillon